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Aligning AIED Systems to Embodied Cognition and Learning Theories

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Abstract. Embodied Learning involves the development of perceptuo-motor schemas as students manipulate physical objects that embody a concept, or when students move in domain-relevant ways, physically enacting solutions to problems, as well as gesturing as they interact with peers face-to-face. Teaching standards (e.g., for K-12 math education) align with such theories of learning. For instance, teachers are encouraged to have students "talk mathematics", construct viable arguments and critique the reasoning of others, model with mathematics, and use tools with precision, among others. Yet most learning technologies and tutoring systems are not designed to facilitate embodied learning; instead, they tend to align with more traditional views of student learning. This paper explores possible ways AIED systems can reflect embodied learning theories, within K-12 STEM education. We present the *WearableLearning* platform as an example that intends to bring embodied learning to mathematics and computing education for students at the K-12 level, with implications and suggestions on how to use Artificial Intelligence to make embodied learning environments more effective.

Keywords: Embodied Learning, Math Games, Mobile Devices, Learning in Motion

1 Introduction

Entering a K-12 classroom and witnessing an experienced math instructor in action is a delightful experience. Within these classrooms, you might see students using tangible tools, such as measurement instruments, number lines, fraction tiles, volumetric objects, and base-10 blocks. These tools are often utilized to enhance conceptual understanding, taking advantage of the physical classroom environment where students engage in activities on tables or on the floor, and participate in rich discussions with their peers and teachers.

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Mathematical objects serve important roles, acting as bridges between familiar, intuitive prior knowledge and new concepts, supporting students to develop abstract mathematical thinking [10][13]. The main role of teachers is to facilitate the bridge from concrete representations to abstract concepts, and this practice continues in good math classrooms throughout secondary school. Alternative methods involve students expressing solutions to math problems through their entire bodies. This may include using the body to measure objects, people, or spaces, and even physically walking out numerical mathematical solutions.

These math teaching practices align with theories about how people learn: Embodied Cognition and Embodied Learning posit that sensory-motor action, including gestures, are essential for teaching and learning, as ideas are distributed among the mind, the world, and the social context, and that hand gestures may support essential communication during learning [19]. Students explore mathematical relationships and concepts by manipulating real objects in the environment and communicating face-to-face with peers and teachers.

Yet, there is a significant disparity between the math learning technologies employed in schools and the effective practices of mathematics educators. Conventional K-12 mathematics educational technologies and tutoring systems are typically not designed to facilitate embodied cognition; instead, they tend to align with more traditional perspectives on student learning. Teaching standards for mathematics have evolved, influencing how math educators are expected to instruct and engage with students. Most traditional learning technologies, however, still resemble digital worksheets. Embracing embodied learning implies moving the Human Computer Interaction (HCI) "off the keyboard", to experience concepts, engage in hands-on and socially-rich activities, and obtain support during moments of struggle and cognitive conflict in real-time.

This article poses how learning technologies might address embodied learning, a shift that entails a focus on refining and deepening interactions with learning technology. For instance, teachers should implement tasks that are hands-on and cognitively demanding, introduce discussion, and require strategic thought and rich problem solving by students [12]. We propose how learning technologies could support learning via the best of two worlds: a) intelligent tutoring systems that assess and trace students' knowledge and affect, and can personalize instruction based on moment-by-moment tracking of students' mental states (i.e., knowledge and affective tracing); and b) Embodied Cognition and multi-modal interaction, encouraging students to physically act on the environment in meaningful ways, for sense making and understanding of ideas.

2 Background Research

Teaching Mathematics. K-12 Teachers are encouraged to employ various representations and tools in the exploration of math ideas, and incorporate the manipulation of various tools, shapes, models, or technology. In the United States, The Math Practice Common Core Standards (MPCCS) encompass all grade levels and emphasize discourse and communication among students in the classroom [11]. Teachers should facilitate discussions, ask appropriate questions, and solicit student

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ideas to support students' mathematical thinking [24], offering ample opportunities for students to "talk mathematics" and share ideas in groups [28]. Discourse involves an explanation of mathematical strategies and students' own justification of their thinking and reasoning [13]. Explanations focus not only on how students obtain solutions but why certain strategies are appropriate.

Traditional Learning Technologies & AIED systems. The affordances of tutoring systems and cognitive tutors as supports to student learning and teacher's pedagogical goals, are fairly well known: a) they provide immediate feedback when students solve problems and enter correct/incorrect answers [31]; b) they provide detailed personalized support when students make mistakes (help provision) or when they seek help during cognitive conflict [3]; c) they enable the use of detailed student logs from student interactions with the computer to provide "stealth assessments" even within educational games, without stopping to test [26]; and d) they allow personal pacing of problem solving work, with students working at their own pace with a "more knowledgeable other" (computer versus human) enabling, to some extent, replication of a reasonable form of a teacher/tutor into as many students in the class, albeit a digital tutor that supports students even emotionally as they learn [8]. Yet, underlying most learning technologies is an information-processing view of learning based on memory, problem solving and practice, where knowledge is considered an internal process in the mind.

Embodied Cognition. The exploration of cognition in research is expanding to encompass a wider viewpoint, with the recognition that it is distributed across the mind, senses, motor capabilities, and social interactions [29]. This theory of embodied cognition assumes that sensory perceptions, motor functions, and sociocultural contexts shape the structure and development of thinking skills, including mathematical thinking. Learning involves the creation, manipulation, and sharing of meaning through bodily interactions [15]. Students may be guided to encounter, discover, and rehearse perceptuo-motor schemas in relation to mathematical concepts and relationships. Research on hand motion gestures while teaching/learning suggests that a motor encoding of our mathematical ideas exists, and thus, that mathematics teaching should use bodily motion, action and hand gesturing [1]. Yet, one problem in embodiment research is the separation between embodied cognition/learning research, and teachers and education practitioners [18].

Frameworks of Embodied Educational Media. Abrahamson et al. [2] proposed a theoretical framework that encapsulates the elements mentioned, offering an ecological-dynamics analysis of learning mathematics through interaction with educational media designed for embodied experiences. This framework includes investigating and informing social practices in teaching and learning of physical movement, which believes in instructional affordances of instructional materials. We naturally ground new concepts in perceived affordances for enacting movement forms that solve motor-control problems. Similarly, the theory of 4E Cognition (embodied, embedded, enacted, and extended) [25], suggests that mathematical ideas are grounded in perceived dynamic images that develop through sensorimotor interaction, and then rise to consciousness through languaging. These theories bear practical implications for how we design learning environments. Students can develop new mathematical knowledge by first learning to move in new ways and only then making sense of this emergent capacity within a discipline --such as mathematics.

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Importantly, we are not claiming that every movement will lead to learning. Certain kinds of movement (e.g. running to the end of the room) might contribute to the engagement/enjoyment, but it is unrelated to learning and not what we are referring to in this paper as "embodiment". Instead, we are referring to activities that involve action, movement and gestures that match, signify, and externalize mathematical ideas. Past research has shown that random movement does not lead to math learning, rather movement encouraged by math-embodied activities has to match, capture, and/or express mathematical concepts to be learned [18].

Embodied Design Frameworks. Principles of embodied design can be used as a critical lens on how digital resources are created and brought to life into mathematics learning tasks. *WearableLearning* facilitates the creation of learning activities that implement themes of embodied interaction design [14]. Embodied games and simulations have utilized a large breadth of design approaches; thus, we rely on a unified design framework [14] that aggregates different conceptual design approaches for embodied learning systems, which is based on: 1) kinds of physical interaction, classified into embodied, enacted, manipulated, surrogate, or augmented; 2) kinds of social interactions that the activity requires (e.g., collaborative, competitive and the role of Non-Player-Characters such as supportive teachers); and 3) role of the environment, and the degree of physical/virtual/mixed world that it involves. We expect teams of teachers and researchers to create activities spanning different areas of this framework, as our existing math game-like activities already cover different areas.

3 A case study: WearableLearning

We present a case study on *WearableLearning* as an example of a learning technology that employs embodied learning theories, both for Mathematics learning and early Computer Science, designed for children in grades 3-8. *WearableLearning* enables K-12 mathematics thinking and learning via physically active and multiplayer math games, using mobile devices (e.g., Wi-Fi-enabled phones or tablets); these are embodied mathematics experiences, quests, measuring objects and space, and even walking out math solutions in physical spaces [4][5]. Students participate in active and collaborative math activities, finding mathematics in their school environments, and using manipulatives and other typical materials in a math classroom (e.g., geometric shapes, measurement tools, fraction tiles). Mobile devices become "virtual assistants," digital tutors that provide instructions and support. Assistants are coordinated across students (for team-based collaboration or cooperation), and multiplex the teacher in as many parts as students, orchestrating activities, providing just-in time help/hints and feedback as they move and "play" by exploring the actual space/objects in real-time.

Mobile devices provide great potential to augment educational contexts and promote tangible, mobile, and ubiquitous learning, as technological devices have become more portable, less cumbersome, and even wearable. Mobile technologies and tools are light, easy to wear, and offer the potential of tracking large amounts of physical activity and individual information. The promise is a new genre of learning technologies that blends with classroom culture, hands-on manipulatives, encouraging sense-making, while still retaining the benefits of intelligent tutoring systems (ITS). *WearableLearning* both promotes human-to-human interaction, and leverages

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technology for what it affords best: self-pacing, just-in-time support, coordination, cooperation.



Figure 1. Children playing *EstimateIT!* in schools and outside, via *WearableLearning*

Game-like activities created by teachers and researchers using the *WearableLearning* platform help grade 5-8 students learn mathematics [4]. These embodied math activities were created for a range of grades and standards, from numbers and operations to algebra to geometry and measurement, for children grades 3-8. The *WearableLearning* Platform and Curriculum have been used to support multi-student math active learning, using mobile devices in real K-12 school settings and afterschool programs. The platform is web-based, supporting embodied developmental mathematics game play for multiple students, blending hands-on activities and collaborative games within the classroom culture. *WearableLearning* supports collaborative, game-based learning that enables students to move about and explore their environment, while guided by mobile devices. It incorporates embodied activities into the realities of classrooms, curricula, and standards [4] and enables students to play social and active learning games that stimulate mathematical thought. It supports multiplayer, interactive, gamified experiences that use smartphones in full classes of K-12 students. While using *WearableLearning*, students can manipulate objects available in the physical environment to take measurements, compute, estimate, and gesture while interacting with their peers and teachers in rich face-to-face discourse. As game players, students use *WearableLearning* through smartphones to solve quests and math problems. As game creators, teachers deepen their math knowledge by designing math games that target the curriculum their students are learning.

Prior research has analyzed the feasibility of teachers, researchers, and students creating physically-active multiplayer math games with *WearableLearning*, and identifies challenges they might encounter [27]. Math game playing and creation with *WearableLearning* has so far been implemented with roughly 500 students, as well as 20 math teachers, who also have created a variety of games for students to play as part of an eight-hour teacher professional development program [28].

WearableLearning activities apply principles of embodied interaction design [14], in that they involve *thinking through doing,* as mathematical reasoning uses bodily capacities; it implements the *bodies matter in performance* principle, as motion/action enable better performance; it implements *high visibility for sharing*, as visible mathematical objects/artifacts support synchronous collaboration. Embodied activities

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also involve *risk-taking*, as a physical action is characterized by risk; choosing an action requires commitment, personal responsibility, and accountability to the team. This results in high levels of affective engagement with tasks, as opposed to technologies that often strive to minimize or eliminate risk.

The promise of the *WearableLearning* technology is a new form of learning environment, augmented by technology, that naturally blends into the classroom to facilitate interactions between teachers, students, and the curriculum, and where collaboration and face-to-face discourse and gesturing between students and teachers is a natural ecological part of the problem-solving and learning process (see Fig 1). A challenge of embodied technologies is maintaining the affordances of embodied interaction while deploying these platforms with full classes of students, at scale.

3.1 WearableLearning and its Math Games Library

The *WearableLearning* Platform enables teachers to play, manage, and create educational game experiences that involve several students, physical movement, and objects in the environment. Mobile devices are worn on the lower arm (as many games require free hands for hand actions, e.g. Fig. 3), yet for other games students may carry them (when the math games involve mostly their legs, e.g. stepping, Fig. 2). Anybody can browse public *WearableLearning* games from its library, to play with students as class-wide activities. WearableLearning users can take on different roles within the platform: Game Players, Managers or Creators.

A math games library is accessible from WearableLearning.org, with their designated Math Common Core Standards (MCCS), which US teachers understand. These games were designed and implemented in *WearableLearning's* Game Editor, and are deployed with its Game Player, fit a variety of Math Common Core standards [11] and student ages/grade-levels, concretely for MCCS areas 3.G, 3.NF, 4.MD, 4.NF, 5.NF, 6.NS, 7.NS., 7.RP, 8.G and 8.EE. These games were created either by researchers and/or by a team of math and STEM teachers during an 8-session workshop. The process of creating games is well understood by now, and an 8-session Professional Development (Smith et al., 2023) already exists to prepare teachers to create math games in *WearableLearning*, via the Game Editor (see Fig. 4). The following are examples of the games available in the *WearableLearning* games' library.

Tangrams Race is a relay-race game designed to teach geometric concepts. Each player within a team wears a smartphone and takes turns to run to an end-line to retrieve geometric shapes whose description is specified in the smartphone. Once students retrieve a geometric piece, they enter a color code that identifies the shape (a sticker on the shape) and receive immediate feedback: if they picked the correct shape, the player runs back with the shape to tag the next teammate. Otherwise, they can ask for hints until they retrieve the correct piece. Once all correct pieces have been retrieved, students work as a team to create a final tangram with their shapes [34].

Estimate IT! is a measurement and estimation scavenger hunt-style game for elementary school children where teams of students search for volumes described in their smartphones in mathematical terms (e.g., "Find a cube with a 6" side"). Players are given an unmarked 12-inch stick to support measurement estimations. Players

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collaborate and discuss geometry concepts present in the game (Fig. 1). The game provides hints about how to measure and/or estimate to find the correct volume (e.g., "Once you find a cube, use the dowel to estimate its height, which should be 6 inches tall"). Once students find the correct volume, they enter an identifier (colored squares sequence, sticker on the volume) into the phone and receive immediate feedback [35].

Integer Hopscotch gives students word problems on addition/subtraction of integers, as players walk along a 20-foot number line to find the answer. Each integer on the number line has a color code, which students input into a mobile device to receive immediate feedback and can request a hint. e.g. about whether they should move right or left. The goal of the game is to work in pairs to complete all the integer problems by "walking along the number line".

What's My Line? is a slope-equation and graph-matching game that involves slope-intercept form of equations to ensure that students can read and interpret equations. Students move through different stations around the classroom and are asked to match equations presented on a smartphone with the corresponding line on a large paper graph where several lines are drawn. Students recreate the correct lines on their own paper pads as they move between stations, before other teams can finish the game. The drawn lines on the team's paper-pad reveal a secret shape.

Flip, Slide, Turn is a game about geometry on a giant coordinate-plane tarp on the floor of a classroom, gym, or a square-tiled patio (Fig 2). Pairs of students share a device—one of the students reads the instructions aloud while the other student becomes a point on the coordinate-plane tarp, and is reflected, translated, or rotated around the origin by walking on the tarp. One student enacts the solution, the other reads aloud and commands (surrogates); students think, argue, celebrate successes [22].

Teachers and researchers have been capable of creating highly innovative math activities with game-like qualities, incorporating various forms of embodiment (enacted, manipulated, surrogate, augmented). The activities also constitute effective tools for math learning, as evidenced by students' consistent improvement in test performance from before to after exposure to these math game-like activities, which encompass diverse forms of embodiment, including the manipulation of objects, enactment of solutions (such as measuring or performing addition/subtraction on a number line), and embodiment through actions like Flip, Slide, Turn, where students represent x-y points.

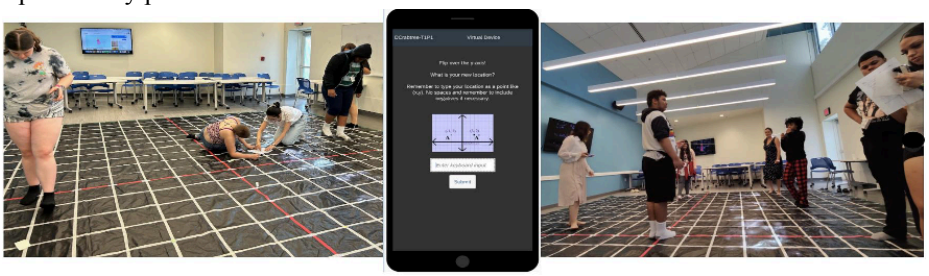


Figure 2. *WearableLearning* utilizes mobile devices to provide teams of students with problem-solving challenges. In the *Flip, Slide, Turn* game, students collaborate and enact the solutions with their full body on a coordinate plane.

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These engaging embodied game-like activities, supported by *WearableLearning*, are the beginning of new research at the intersection of problem solving, embodied action and gesture, and mobile learning technologies. The games that teachers and researchers created were implemented in real math classroom settings and showed improvement in math problem solving performance across different areas of math and grade levels, even after 30–40-minute exposures. All games have led to significant learning gains so far, comparing pretests to posttests, and while the results presented don't explicitly reveal it, students were observed to be incredibly engaged while playing these games.

The kinds and degree of embodiment varied with each math game, suggesting that future research and development should focus on understanding how physical actions in each game mediates student learning and the affordances of the materials, objects, space, and human interactions, as mediators of math learning. Teacher-created games appeared to be more effective than researcher-only created games in general, probably because teachers have more pedagogical and classroom knowledge, which allowed them to target specific areas where students needed improvement, and created tests that better assessed the material taught by each game.

4 Research Line #1: Analyzing the benefit of Embodiment in LTs

Similar to much research on effective instructional strategies within Learning Technologies [3][24], an important line of research relates to understanding the potential of embodiment (domain-relevant motor action and gesturing) compared to more traditional stationary learning technologies, and compared to business as usual in the classroom (e.g. math class using manipulatives but without technology). For instance, initial research studies by the authors have attempted to understand the potential of the embodied *WearableLearning* concept, in comparison to control groups. One of the studies compared a classroom lecture to the *Tangrams Race*, implemented via a prototype of *WearableLearning*, and yielded a medium effect size for math improvement, where an experimental *Tangrams Race* condition outperformed a control business as usual condition [34].

A second study analyzed the potential contribution of the affordances of the physically active *TangramsRace*, compared to a stagnant one. The experimental condition was *TangramsRace* in *WearableLearning*, as shown in Fig. 3 The control condition was a digital platform that provided the same questions and the same hints, but the it required students to sit down and solve questions on a traditional math practice technology, with immediate feedback, and with hints available upon request, similarly to the Tangrams Race in *WearableLearning* (experimental condition). Participants were N=54 students from grades 3-5 in afterschool programs in the United States, who were randomly assigned to the experimental (*Tangrams Race*) or sitting down (control) condition. The study yielded no significant results ($p<0.17$), yet, the effect size corresponded to a medium effect size, Cohen's $d=0.36$, favoring the Tangrams Race condition.

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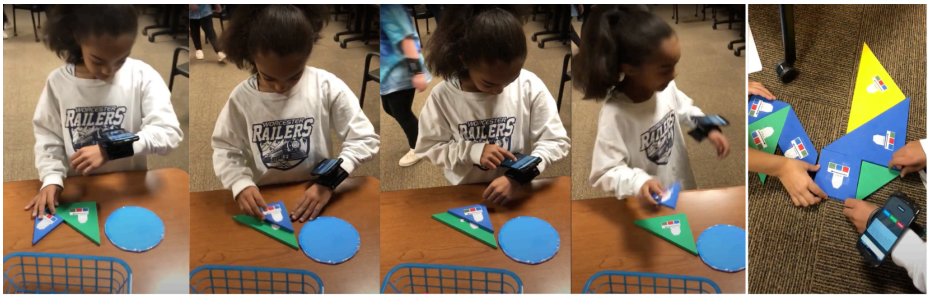
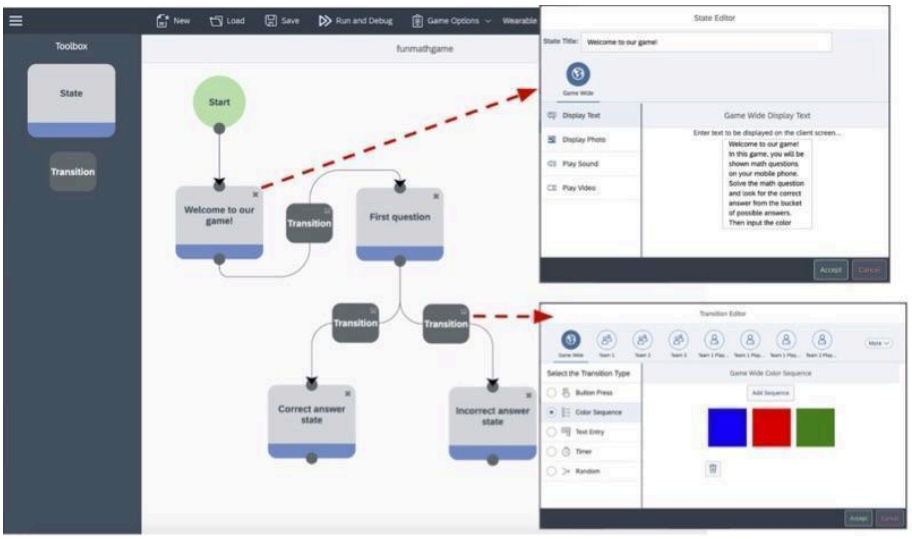


Figure 3. A student playing the *Tangrams Race game*. From left to right: a) Student reads the mathematical constraints of the sought object; b) student rotates triangle and compares sizes using hands; c) student enters color code for identifying object; d) after receiving feedback, the student picks the object and runs back; e) at the start line, students collaborate to build the requested Tangram.

Embodied Hints and Support. Researchers have analyzed hand gestures as part of STEM lectures to improve concept understanding [9]. Given the embodied nature of our learning tasks, i.e., actions to succeed in the math games are highly physical, gesture-based hints in embodied tutoring systems should allow struggling students to receive improved support while playing games. In one research study [7] we performed an *embodied cognitive task analysis* of children's gestures/actions during measurement tasks. Using data from that study, the authors created hints that conveyed the most common motor strategies observed from college students, which acknowledged the most common mistakes of elementary students. A controlled study analyzed the benefit of having "embodied hints" on the phones [35], which tapped on those typical estimation/measurement errors. This study yielded a medium effect size of Cohen's d=0.37, in favor of the "embodied hints" condition. Given the results we consider that the comparison of carefully designed embodied games, deployed via *WearableLearning*, and the creation of digital embodied support, has good promise of yielding a significant benefit compared to business as usual in future studies.



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Figure 4. Authoring Games as Finite-State Automata: the *Game Editor* in WearableLearning

Authoring Embodied Games w/ narratives. Embodied tutoring systems need an authoring environment to create content for it. In *WearableLearning*, the Game Editor environment is designed to create/program unique games with a wide variety of content, levels, and modes of collaborative or competitive play, for players or teams to engage in. As Game Editors, students define games as Finite State Machines (FSMs) that describe the behavior of players' mobile devices (see Fig. 4). FSMs are computational models that can be used to simulate phenomena, sequential logic, and the behavior of computer programs. They include a finite number of states and directional transitions between states (including loops). Game Editors program their games using *WearableLearning's* drag-and-drop FSM-based programming language, where they specify output states to indicate media to display on mobile screens (e.g., text, images, videos) for the purpose of providing game instructions, questions, or hints, that involve interacting with objects and the environment. These instructions in the phone follow a logic towards a narrative in the game, which can vary per game. For instance, in the *Tangrams Race*, the game needs to guarantee that all seven pieces required for the final puzzle are collected by different students at the end line, for them to succeed at putting the puzzle together in the end. Thus, there are constraints about objects that need to be retrieved by children, that guide them to find, manipulate and retrieve correct objects.

States correspond to *screen outputs* and can be configured to display different media to make interesting prompts such as displaying a math problem with pictures related to the problem, combined with an audio recording. Game Editors also specify transitions between states to indicate a possible input that players can submit (e.g., button presses, text input, color sequences that identify objects) to move from one state to the next (see Fig. 4). Transition input specification enables different game response interactions; for example, game creators can ask players to respond to problems with a text entry (e.g., "How many sides does a triangle have?), or use buttons to enter choices (e.g., "Push the black button if you want a hint"), or type color sequences that identify objects (e.g. red, blue, green may identify a certain kind of object to be sought for). Game Editors can test-run and debug their games in the built-in debugger, which simulates gameplay by launching a phone simulator to play-test each game and player's actions.

Students creating physically active games develop Computational Thinking. WearableLearning allows teachers and students to visually program and debug games via a drag-and-drop programming interface to specify the behavior of game players' mobile devices that support players throughout the games. The WearableLearning curriculum [8] was created to entice teachers (PD) and their students to engage in game design and creation. Designed for programming novices, 5th-10th grade students have created games by designing them first on paper and later within the Game Editor. Research studies have suggested that teachers develop CT as they create math games [27]. Yet, it is not only teachers that have created games, but also several research studies have explored how students creating games helps them develop STEM skills by creating the games themselves. The *WearableLearning* math game creation curriculum has been used by ~25 teachers and 100 students, and has

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suggested that this 7 hour activity enables them to gain computational thinking skills, particularly algorithmic thinking and knowledge about models and simulations [21].

5 Research Line #2. Enhancing Embodied Technologies with AI

The *WearableLearning* game-based environment operates as a traditional tutoring system, however, this also incorporates bodily movement and interaction with objects, peers, and the physical space. In *WearableLearning*, there are also correct and incorrect answers, for chosen objects in the real world (or for configurations of objects). There are also hints that can be requested. Similarly to game-based tutoring systems, there is a narrative that needs to be followed temporally, and occasionally there are chance actions carried out (player choices).

Assessment of Student Knowledge via Multimodal Learning Analytics. The goal of Learning Analytics is the study of the low-level traces left by the learning process, in order to better understand and estimate one or more learning constructs that are part of the process and, through carefully designed tools [32]. Most AIED systems, including *WearableLearning*, collect detailed log data from students, math-task performance in real-time as students carry on tasks and interact with the technology. A new line of research involves the possibility to feed formative assessments back to teachers after students work with physical objects, or with their full bodies, using sensors for assessment of students' motion and gesture. Teachers spend too much time on assessment (usually teaching, pausing to give tests, and then re-teaching); this takes time away from learning opportunities. *WearableLearning* gathers large amounts of data from students while they work without pause. This enables the possibility of providing motor "stealth assessments" [26] as students act/interact with mobile devices as part of participating in mobile game-like activities.

Like any game-based AIED system, it is possible to estimate students' knowledge while playing a game-like activity. In *WearableLearning*, each "state" can contain meta-data, such as the fact that it is a "math question", and also the mathematics "skill (standard)" that is associated with. Other states are "answer" states and may be categorized as "correct" or "incorrect" states. Succeeding or failing at the math question (by arriving at the expected correct answer state, that may correspond to entering a code that indicates a correct/incorrect object in the environment) is evidence of a student's knowledge. This means that knowledge can be modeled with a traditional Bayesian Estimation of students' knowledge based on correctness, errors, and hint requests [26][33], even in these mobile game-based embodied environments. While *WearableLearning* does not yet make such assessments for each student), that is the long-term vision for this technology, especially the possibility of sensing specific motor actions/gestures automatically, when motion sensors are worn.

Fig. 5 shows new possibilities of AIED research where lower-level sensor data can help support higher levels behaviors and motor actions first, to later infer physical gestures and actions, which should help assess higher level cognitive/affective constructs, as well as domain-level practices (e.g. mathematics common core practices that are encouraged in a mathematics classroom). The example in Fig. 5 shows that a variety of hand movements, as well as full-body movements, can be inferred from

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sensors that are worn or carried, either on mobile phones/watches, or via ring/bracelet sensors on the hand. These behaviors, in combination with typical HCI behaviors, can be used to recreate motor gestures and actions that are domain relevant. These, in turn, can be used to infer cognitive/affective states, and even Mathematics Common Core Practices. Creating models that can make these inferences involves collecting data from real students as they engage in these embodied activities in schools, gathering sensor data using sensors, and also videotaping students, to later annotate a variety of motor actions/gestures in videos, as well as cognitive/affective student states. By fusing, triangulating and integrating these sources of data we can obtain an invaluable datasource that captures individual and collaborative behaviors, affective/cognitive states, gestures, actions and interactions with technology and real objects, math strategies, as well as kinematic descriptors and motor data, at a high level of detail. Educational Data Mining and Machine Learning can be used to make these higher-level inferences from lower-level sensorial data streams. Once created, these models can be used to make assessments, which can support not only basic research on how people learn, but also support teachers through Dashboards (under development) to track students' progress and provide formative assessments [20], without stopping to test.

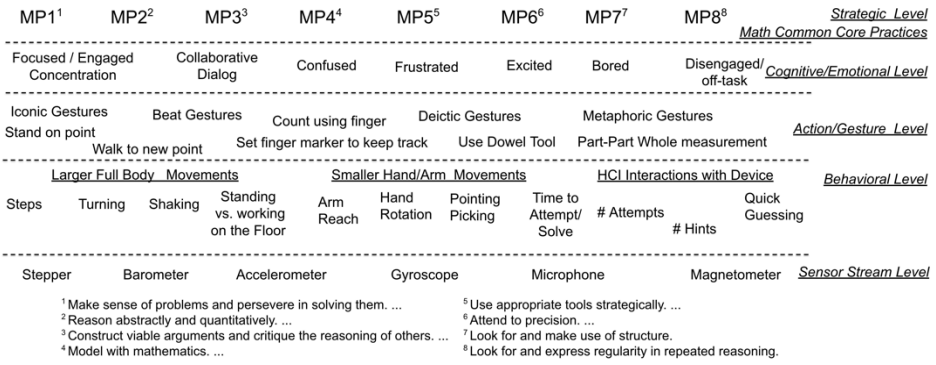


Figure 5. Inferring Mathematics Common Core Practices [11] from sensor data

3 Discussion

During the COVID pandemic, teachers and parents felt children spent "too much time" on the computer, resulting in a negative impact on social skills development and human communication [17]. While some tutoring systems may allow collaborative learning via chat, there is something clearly missing regarding the benefits of face-to-face discourse and gesturing in education. Embodied Learning involves collaboration and gesturing with other humans. There is much to study about the benefits of face-to-face student interaction and sharing of spaces, mediated by physical and digital objects, in comparison to chat-based communication, or traditional digital learning environments.

The main benefits of Intelligent Tutoring Systems regard smart pedagogical decisions based on a moment-to-moment automatic assessment and understanding of the student at various levels. In this BlueSky paper, we have argued that it is possible

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and feasible to create AIED systems that better fit the reality and needs of K-12 classrooms, by aligning to Embodied Learning principles. While embodied learning research is pinning down the precise cognitive role of motor action in conceptual learning [1], the objective of digital resources should be to foster opportunities for students to develop new situated perceptions, mobilizing the prospective enactment of mathematical ideas and mathematical transformations [2]. This implies the need to create novel technologies that enable the enactment of mathematical practices, while at the same time automatically assessing how students move, and the impact these actions/gestures have on cognition, affect and learning.

The idea of using sensors for assessments of affective/cognitive constructs is not new [6]. The main difference with prior research is two-fold: a) the fidelity that embodied learning technologies can bring to the reality of teaching/learning experiences of K-12 classrooms; b) the growing importance of motion, action and gestures in human learning; c) the opportunity, given abundance of sensors in mobile (and wearable) digital devices. Yet, for a meaningful impact, it is essential that embodied digital learning technologies efficiently scale, to meet the demands of real-world settings, including schools and classes with a ratio of 20 students per teacher.

Novel multimodal learning and assessment systems cannot assume that students will always be ready and "in the mood" for learning, nor have all the prerequisite knowledge available to them. Yet this is the strong point of intelligent tutoring systems and other AIED systems, which can respond to students in a variety of ways depending on a variety of cognitive/affective states. There are many opportunities ahead at the intersection of AIED and embodied learning, in the K-12 education arena.

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